Revision Guide for CPT107 and Past Exam Papers

Sample Exam Papers

- ***To obtain the full marks for each question, relevant and clear steps should be included in the answers.***

- ***Partial marks may be awarded depending on the degree of completeness and clarity.***

Exam Papers: Proof by Contradiction

Let $x \in \mathbb{Z}$. If x^2 is even, then x is even. If you think that it is true, prove it. Otherwise, give a counterexample.

Answer:

True.

Proof by contradiction -

Step 1: Suppose x^2 is even, then x is NOT even (the assumption).

Step 2: There exists an integer y such that $x = 2y + 1$

Step 3: Then $x^2 = (2y + 1)^2 = 4y^2 + 4y + 1 = 2(2y^2 + 2y) + 1$, so x^2 is odd.

Step 4: This is a contradiction, so the assumption must be wrong.

Exam Papers: Proof by Induction

For every integer $n \geq 1$, use proof by induction to show that $5^n - 1$ is divisible by 4.

Answer:

Proof by induction -

Base Case: n=1, i.e., 5^1 -1 = 4=4× 1

Inductive Step: Assume it is true for n=k for $5^k - 1 = 4r$ where r is an integer

Proof continues for n=k+1:

$$5^{k+1} - 1 = 5 \times 5^k - 1 = 5 \times (4r + 1) - 1 = 4$$

Exam Papers: Set Theory

Let the universal set be $\mathbb{N}$, $A = \{x \in \mathbb{N} \mid x \text{ is even and } 1 < x < 10\}$, $B = \{x \in \mathbb{N} \mid x \text{ is odd and } 3 < x < 9\}$ and $C = \{x \in \mathbb{N} \mid x \text{ is odd and } 2 < x < 8\}$. Find the elements of:

1. $A \cup (B \cap C)$
2. $A \cup (B - C)$
3. $(B \cap \sim A) \cap (A \Delta C)$

Answer:

A={2,4,6,8}, B={5,7} and C={3,5,7}

Then $(B \cap C) = \{5,7\}$

$A \cup (B \cap C)$……

Exam Papers: Functions

Let A, B and C be sets and let $f: A \to B$ and g:B $\to C$ be functions. Prove that:

1. if $g \circ f$ is injective, then f is injective
2. if $g \circ f$ is surjective, then f is surjective
3. if $g \circ f$ is bijective, then f is injective and g is surjective

Answer:

1. Suppose that $g \circ f$ is injective. Let $x, y \in A$, and suppose that f(x) = f(y). Then g(f(x)) = g(f(y)), so $(g \circ f)$(x) = $(g \circ f)$(y). Since $g \circ f$ is injective, it follows that x=y, which proves that f is injective.
2. Suppose that $g \circ f$ is surjective, and let y ∈ C. Then c = $(g \circ f)$(x) for some x ∈ A. Then f(x) ∈ B and g(f(x)) = y, which proves that g is surjective.
3. Since $g \circ f$ is bijective, $g \circ f$ is both injective and surjective. The result now follows from the previous two proofs.

Exam Papers: Functions

Let $f: \mathbb{R} \to \mathbb{R}$ be given by $f(x) = 3x^2$, give two reasons why $f(x)$ is not invertible.

Answer:
For instance, f(10)=f(-10) and there is no real number x such that f(x) = -20. These two examples shows that f(x) is neither injective nor surjective, and we can conclude that f(x) is not invertible because a function is invertible if and only if it is a bijection.

Exam Papers: Logic

Show whether the following statement on the left-hand side is equivalent to the statement on the right-hand side using logical equivalences:

$$(p \to q) \vee (p \to r) \equiv p \to (q \vee r)$$

Answer:

$$\begin{aligned}
(p \to q) \vee (p \to r) &\equiv (\neg p \vee q) \vee (p \to r) && \text{Logical equivalence using conditionals} \\
&\equiv (\neg p \vee q) \vee (\neg p \vee r) && \text{Logical equivalence using conditionals} \\
&\equiv (\neg p \vee \neg p) \vee (q \vee r) && \text{Associative and Commutative Laws} \\
&\equiv \neg p \vee (q \vee r) && \text{Idempotent law} \\
&\equiv p \to (q \vee r) && \text{Logical equivalence using conditionals.}
\end{aligned}$$

Exam Papers: Pigeonhole Principle

XJTLU ICS programme has 500 students. Show that at least two of them were born on the same day of the year.

Answer:
There are 500 students (pigeons), and only 365 days they could have been born on (pigeonholes), so by the Pigeonhole Principle, some two students were born on the same day.

Exam Papers: Probability

An urn contains 12 red balls, 5 green balls, and 8 blue balls. You pick 6 balls at random (without replacement). What is the probability that you get 4 red balls, 1 green ball, and 1 blue ball?

Answer:
There are 25 balls, and you pick 6; this can be done in (25, 6) ways. We must pick 12 red balls, which can be done in (12, 4) ways; similarly, we must pick 1 green ball in one of (5, 1) ways, and 1 blue ball in one of (8, 1) ways. Hence the probability of getting 4 red balls, 1 green ball, and 1 blue ball is:

(1/(25, 6))*(12, 4)*(5, 1)*(8, 1)